

Micro-Camera based Airotor

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Yeshwantrao Chavan College of Engineering

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In partial fulfillment of the requirement

for the award of the degree

Of

Bachelor of Engineering in Electronics Engineering

By

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Under the guidance of

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CERTIFICATE OF APPROVAL

Certified that the project report entitled “**Micro-Camera based Airotor**” has been successfully completed by **Sushant Deo, Bhushan Khandekar, Himanshi Tekade Aishwarya Petkar, Rutuja Jambhulkar** under the guidance of **Dr. Yogesh Suryawanshi** of in recognition to the partial fulfillment for the award of the degree of B.E. Electronics Engineering, **Yeshwantrao Chavan College of Engineering (An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)**, Nagpur during session 2021-22

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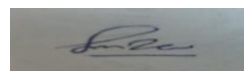
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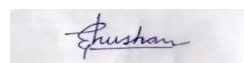
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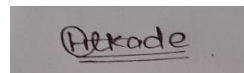
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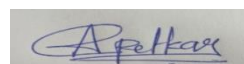
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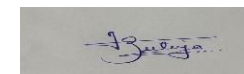
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ABSTRACT

Technological advancement in the field of dentistry have reformed the concept of photography as powerful medium of expression and communication. It helps in revelation of the hidden and overlooked defect in teeth and other part of the cavity.

The Present system invention providing a Micro-camera and LED attachment for dental Airotor/Hand-piece. The Micro-camera viewing attachment provides lens that magnifies the cavity preparation site, LED which will help illuminate the site.

The system significantly improves a dentist's ability to assess complete removal of decay and proximity to patient nerve.

The system can also be used in performing root canal. In general Airotor/Hand-piece are used for various purposes such as oral cleaning, removal of tooth decay, during treatment and with the advance of technology we have introduced an imaging airotor/hand-piece having LED to it.

The system of the present invention has now many advantages because camera attachment incorporates an LED which increases the amount of available light on cavity preparation area.

In order to use such an imaging hand-piece or airotor dentists can also discontinue working on patients set up and acquisition mode for image to be taken identify image to be acquire and its angulation.

Identify what is captured and its orientation and when if needed continue working in patients.

CHAPTER 1:- INTRODUCTION

1.1 OVERVIEW

A dental airotor or hand-piece is a hand-held, mechanical instrument used to perform a variety of common dental procedures, including removing decay, polishing fillings, performing cosmetic dentistry, and altering prostheses.

The airotor itself consists of internal mechanical parts which performs a rotational force and provide power to the cutting tip, usually a dental burr. The type of apparatus is used to perform various dental operations.

It is common for a LED and a cooling water-spray system to also be incorporated into certain airotor. But in a dental airotor, treatment may take longer amount of time to perform the task due to limitations of viewing angles. So by using micro-camera in a dental airotor we can reduce the limitations of viewing angles and it will also help to perform treatment with much accuracy, visibility and overall success of the procedure.

Removable Micro-camera based dental hand-piece such dental hand-piece comes with removable camera attachment so that it can easily detached from dental hand-piece. The camera is attached to outer surface of the dental hand-piece.

Due to attachment and detachment system, it is easy for dentist to sanitized or sterilized both camera and hand-piece. Micro-camera used in this project is equipped with tangible shape with in built LED and a 720p HD resolution.

1.2 LITRATURE SURVEY

In recent years, various invention and improvements have been made in the field of dental instruments to help dentists to fill cavities. Handheld devices with mirrored surfaces provide visible access of treatment site of mouth to dentist, but not all parts of the mouth can be view directly. For example, the dental mirror attached with dental drill hand-piece. This dental drill hand-piece with mirror attachment could only viewed the site of drilling being performed. This mounted mirror is beyond from the dental drill, interfering with the other teeth and also it limits to small images only. So, in this we talked about the dental instrument with video recording device and light source which helps the dentist to view the treatment site of the interior of patient's mouth on display means during treatment without interruption of switching the acquisition mode to check the state of treatment site.[10]

In a present time, dentists used intra oral camera's for live recording or photographing of the patient's mouth, which is before and after the operations or treatment such as removing decay, polishing fillings, performing cosmetic dentistry, altering prostheses, etc. Such system used separate photographing camera for charting records. In order to use such an imaging hand-piece, a dentist must discontinue working on a patient, dentist needs to take the image before the treatment, then identify the image to be acquired and its angulation, identify what is captured and its orientation, then, if needed, continue working on the patient, repeating the process, as needed, in order to complete the imaging sequence for the treatment procedure, it is needed to capture the image after the treatment for ensuring complete removal of decay. There are limitations to such system which includes: the amount of time for treatment and photographing, having interrupt the treatment by photographing the sites of treatment, sometimes it is difficult to open injuries as location the of image orientation. This hand-piece comes with removable camera attachment and low speed hand-piece. The camera attachment made up of silicon which is encased camera board with lens. The camera consists of LED which illuminates the site and visual information transmitted to display device. As it is low speed dental hand-piece, it means it does not use water and provides clear image. It is used during operation of treatment site as it provides real time live recording of treatment site during operation. The camera attachment is located external to the hand-piece, it can be easily removed, cleaned, sterilized, and replaced. [1]

The use of handheld dental instruments such as drills and scales, are itself block the treatment site. Although mirrors are regularly used by dentist to prevent this situation, also using free hand of dentist is annoying and difficult to interpret due to image reversal. Then to eliminate the need for handheld mirrors, the dental professionals provides the camera for intra-oral use. Then Oral cavity camera used in the field of dentistry to capture the image of the site before, during and after the treatment. In this paper, the dental instrument with camera device provided with camera components such as led, image capture, processing and transmission integrated into hand-piece body with easy attachment and detachment system for sterilization of dental instrument. This instrument use imaging chip to remove the need of fibre optics, also using miniaturized camera would reduce the size of camera device assembly of components.[9]

In general, airtor are used for various purposes such as vacuum suction, implantation, tooth decay, plug removal, oral cleaning etc The traditional dental hand-piece uncomfortable and inconvenient to perform treatment while looking directly at inside the patient's mouth and the traditional oral camera can only used before and after the treatment to confirm the state of affected area. Capturing photos of treatment sites or transmitting photos to display device.[2]

1.3 MOTIVATION & PROBLEM STATEMENT

In the field of dental treatment when carious tooth/ cavity or oral health problem occurs it is treated with the help of cutting and removing decay operator i.e dental airotor\hand-piece.

Nowadays dentists are seeing inner view of patient mouth with the help of small mirror And treat patients accordingly dentists have to be more careful they have to take more precautions while treating patients as there is no source of light and accuracy things.

Limitations of such system include amount of time the process take therefore such a system.

Require a separate imaging micro-camera that is going to take image, recording and display it

On screen with advancement of technology we are going to introduce the camera viewing attachment include LED that illuminates cavity preparation site so that dentists can easily see the inner view of mouth and treat patients.

1.4 OBJECTIVE

THESIS OBJECTIVE

- To study the medical field method used for dentistry purpose.
- The photographic method produced higher reliability levels than direct clinical method.
- For the accuracy purpose while using dentistry tool (airotor / hand-piece)
- Validation of result by using appropriate tool(micro-camera based airotor)

1.5 THESIS CONTRIBUTION

The purpose of a literature review is to:-

- Provide foundation of knowledge on topic.
- To identify areas of prior scholarship to prevent duplication and give credit to other researchers
- To identify inconsistencies: gaps in research, conflicts in previous studies, open questions left from other research.
- To identify need for additional research (justifying your research).
- To identify the relationship of works in context of its contribution to the topic and to other works.
- To place your own research within the context of existing literature making a case for why further study is needed.

CHAPTER 2:- REVIEW OF LITERATURE

REVIEW OF LITERATURE

Takeshi Hasegawa, Michizo Yamanaka, “Headpiece with built-in camera and dental treatment method using said hand-piece”, explains the dental hand-piece with built in camera, the dentist perform operations as well as simultaneously can also watch live clear colour internal view of patient’s mouth on display device by using camera fixed inside hand-piece. Camera system consists of colour image sensor CMOS used to provide colour image signal that is transmitted to display on monitor and condensing lens unit with focus range of 3-50 mm for wide clear colour image of treatment site, it means high practical value can be provided.[4]

Hartmut Braetsch, Michael Grimm, “Dental hand-piece with a device for image acquisition”, elaborates that the dental hand-piece with a device for image recording, consisting of an image guide and a light guide are built into the hand-piece, This provides a minimal power requirement and increased image quality due to short-end cable routes, to ensure the cleaning, disinfection and cleaning, to enable independent energy supply, it can also documented the treatment process. For light guide or image guide this uses a prism or angled mirror.[5]

Hartmut Braetsch, Michael Grimm, “Dental hand-piece with device for taking pictures”, this shows that a hand-held instrument for use by dental surgeons, it has a probe housing a conductor to direct a light source output onto a mirror built into the tip. The video image is transmitted through an optical route to be received by a video camera via a lens. A tube includes water and air lines and an electrical lead transmitting the camera output to a processing unit that includes a VDU, printer and memory system. ADVANTAGE - Provides video information as diagnostic aid.[6]

안민석, “dental treatment instrument combined with micro optic device and display system”, this invention explains that the hand-piece consists the micro-camera device which is inserted into the oral cavity of the patient to photograph and display the intraoral procedure site, ; micro-camera used for capturing image information of an intraoral treatment site. Camera detachably attached to an outer surface of the hand-piece is connected to the photographing means. Transmit an image signal photographed by the photographing means; Diffraction means provided between the photographing means and the connecting means used enable the photographing means. Camera device including a control means for receiving a user command to control the photographing means and the diffraction means and to store an image signal transmitted through the connection means and display device connected to the micro camera device to display image information received from the micro camera device on a screen in real time. [7]

2.1 OVERVIEW

Removable Micro-camera based dental hand-piece such dental hand-piece comes with removable camera attachment so that it can easily detached from dental hand-piece. The camera is attached to outer surface of the dental hand-piece. Due to attachment and detachment system, it is easy for dentist to sanitized or sterilized both camera and hand-piece.

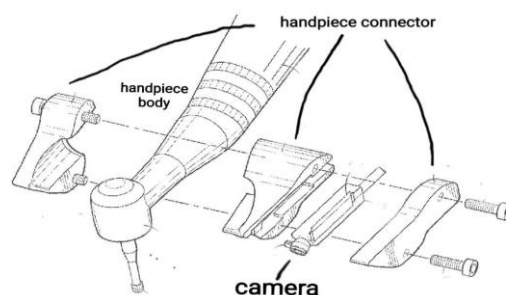


Figure 2.1.1 Hand-Piece Body

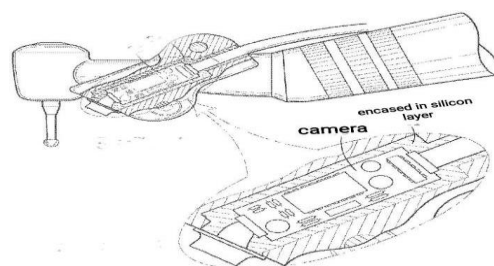


Figure 2.1.2 Hand-Piece with camera

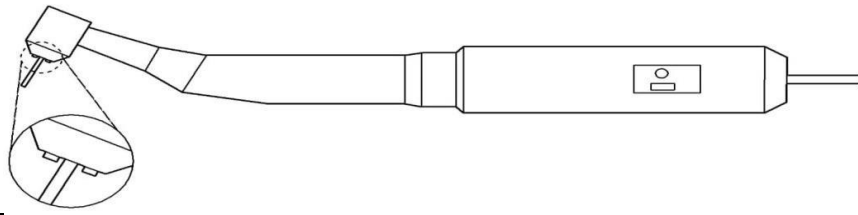


Figure 2.1.3 Simple Airotor

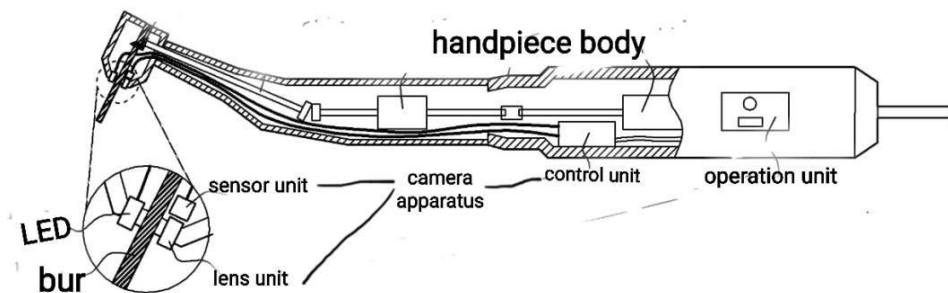


Figure 2.1.4 Airotor having LED and Micro-Camera

Drawbacks

Since, if the hand-piece based on integrated micro camera then it has complex structure. Also, it is difficult for the dentist to sterilizing the hand-piece.

As the high speed hand-piece used water spray, because of it fogging arise and camera view blocked by fogging. Therefore, it is better to use low speed hand-piece as it does not used water spray.

CHAPTER 3:- WORK DONE

3.1 Experimental Apparatus and Methods

3.1.1. Test Air Turbine Hand-piece

Figure 3.1.1 shows the test air turbine hand-piece and Table 1 shows its specifications. In addition, Figure 3.1.2 shows the schematic of the turbine section. The turbine section, which forms the head section of the hand-piece, comprises a nozzle, rotor, and casing and is a single-stage impulse turbine. Figure 3.1.3 shows the rotor, while Figure 4 shows its schematic. This air turbine hand-piece jets out the compressed air that enters from the entrance through the nozzle, and the air is driven as the jet works on the rotor. The inner diameter of the nozzle is $d = 1.3$ mm, and there is one nozzle. The outer diameter of the rotor is $D_1 = 8.8$ mm, while the inner diameter is $D_2 = 3.6$ mm and the blade width is $b = 3.3$ mm, of which there are eight blades.

nozzle	D:nozzle inner diameter[mm]	1.3
	Number of nozzles1	1
Rotor	rotor outer diameter [mm]	8.8
	rotor inner diameter [mm]	3.6
	blade width [mm]	3.3
	blade angle [°]	90
	Number of blades	8
Casing	casing inner diameter [mm]	9.2
	casing width [mm]	4.5

Table 1. Specification of turbine section



Figure 3.1.1 Air turbine hand-piece

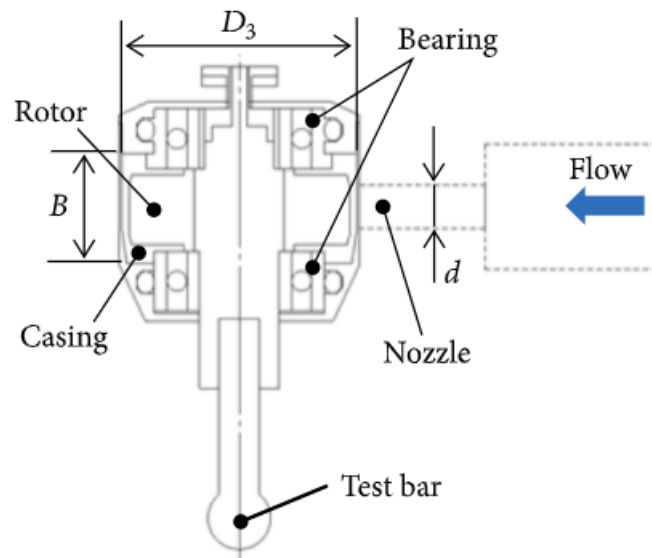


Figure 3.1.2 Schematic of turbine section

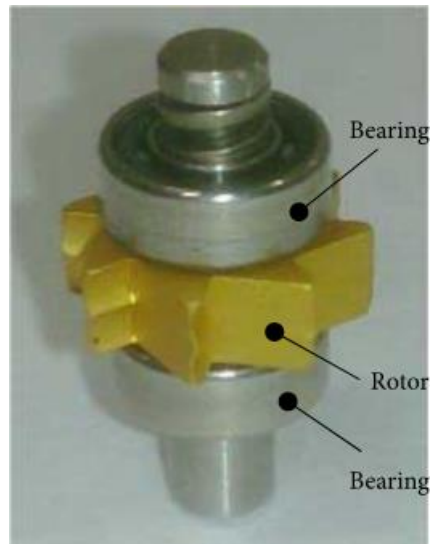


Figure 3.1.3 Rotor

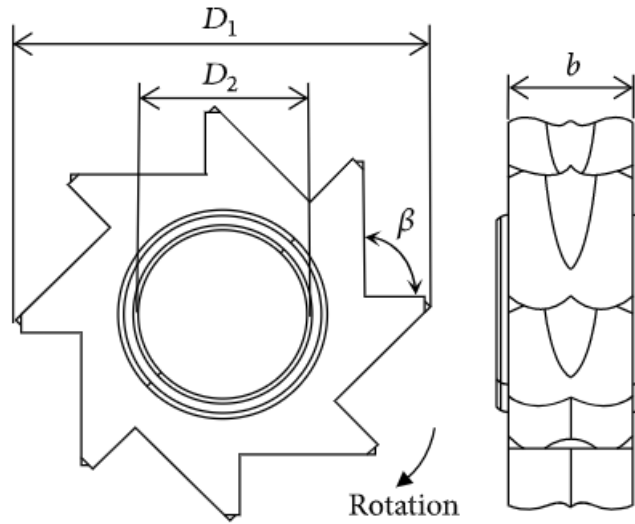


Figure 3.1.4 Schematic of Rotor

3.2 Experimental Apparatus and Methods

The rotation performance of the air turbine hand-piece is assumed to be equivalent to turbine performance, and there are two methods to measure this value. One is when a test bar is loaded onto a work material at a constant speed, and the other is when the test bar is loaded onto a work material at a constant load [2]. Experimental results from the constant speed load method with the descending speed of 1 mm/min are consistent with the constant load method [2]. Thus, in this study, we used the constant speed load method, which has a relatively easy mechanism and makes experiments simpler. The experimental apparatus is shown in Figure 3.1.5 and its schematic is shown in Figure 3.1.6. An air turbine hand-piece with a test bar (cemented carbide with radius $r=1$ mm) attached to the rotational axis of the rotor is fixed in the movable part and is lowered at a constant speed. Compressed air of 0.22 Mpa is injected through the inlet of the hand-piece, and when the rotor is turned on, the test bar is rotated. At this time, the pressure was adjusted according to the regulator (SMC Corporation; AW2000) installed at the hand-piece inlet, and measured using a pressure gauge (Asahi Gauge Manufacturing Co., Ltd.; 101-F510, accuracy $\pm 1.6\%$ of full scale). The test bar is lowered at a descending speed of 1 mm/min, and the freely rotating test bar is pressed against the work material (cemented carbide) on the support stage. The horizontal force at this time F_H and rotational speed n were continuously measured with the load cell (Kyowa Electronic Instruments Co., Ltd.; LMB-A-100N, nonlinearity $\pm 0.5\%$ rated output) and fiber sensor (KEYENCE CORPORATION; FU-20/FS-N11N). The torque T and turbine output L were then obtained with the following equations using the horizontal force F_H and rotational n speed that were measured:

$$T = \frac{F_H r}{1000}$$

$$L = \frac{2\pi n T}{60}$$

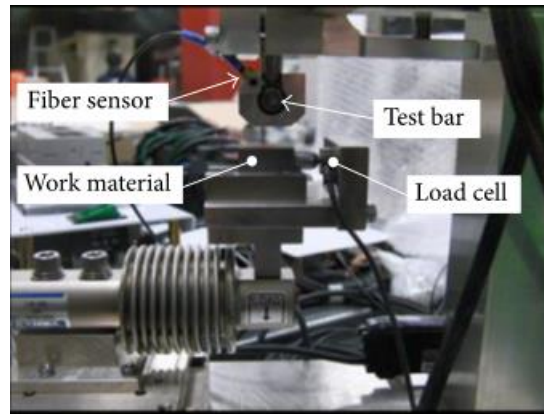


Figure 3.1.5 Experimental apparatus

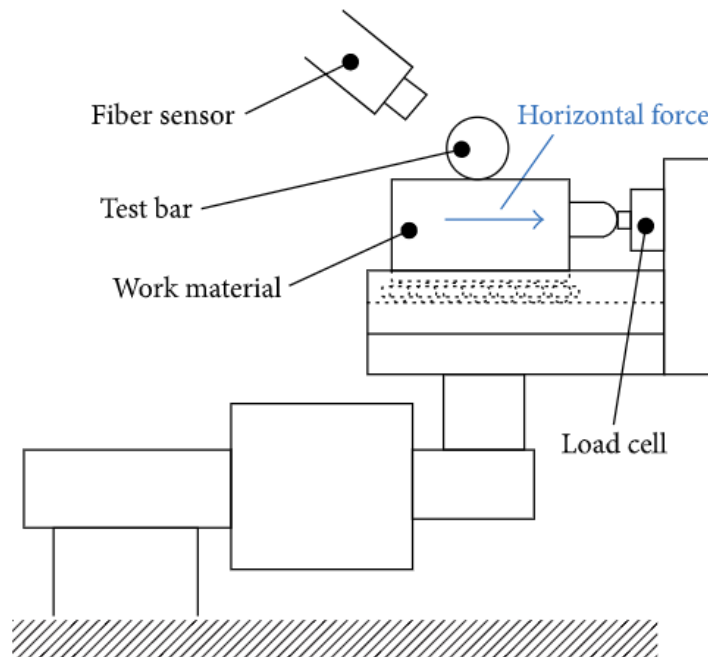


Figure 3.1.6 Schematic of experimental apparatus

3.3. Numerical Analysis Method and Conditions

In this study, we performed a 3D compressible unsteady flow analysis by modeling the whole air turbine hand-piece. For the analysis, we used general-purpose thermal fluid analysis software, ANSYS CFX14.5 (ANSYS, Inc.), and the working fluid was air. The governing equations were the mass conservation equation, momentum conservation equation, energy conservation equation, and ideal gas law [15], which were solved with the finite volume method. Furthermore, we used the Shear Stress Transport (SST) model [16] as the turbulence model. Figure 7 shows the computational model for the whole air turbine hand-piece, and Figure 8 shows the turbine section region. The calculation regions are inlet flow region, turbine section region, and outlet flow region.

The numbers of computational elements were approximately 391,000, 1,127,000, and 783,000 for the inlet flow region, the turbine section region, and the outlet flow region, respectively, corresponding to a total number of elements of approximately 2,301,000. Boundary conditions were a total pressure of 0.22 MPa and static temperature of 25°C for the inlet boundary and static pressure of 0 Pa for the outlet boundary. In addition, we provided arbitrary rotation speed to the rotor region, and the wall surface had adiabatic and nonslip conditions. The boundary between the rotational region and static region was attached using the Transient Rotor-Stator [17]. Time step size was determined the rotor rotates once with 120 steps at any speed. Calculations were continued until fluctuations in the torque stabilized. In terms of grid dependency, the number of computational elements was increased approximately 1.52 times, and it was analyzed with a rotation rate of $n=250,000 \text{ min}^{-1}$. The result was that this decreased by approximately 2.2% in relation to the torque in this paper, and it was confirmed that the effect of the number of computational elements was comparatively small.

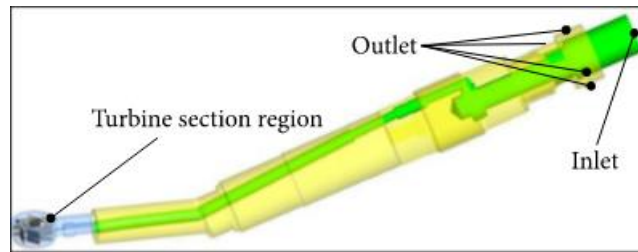


Figure 3.3.1 Air turbine hand-piece model

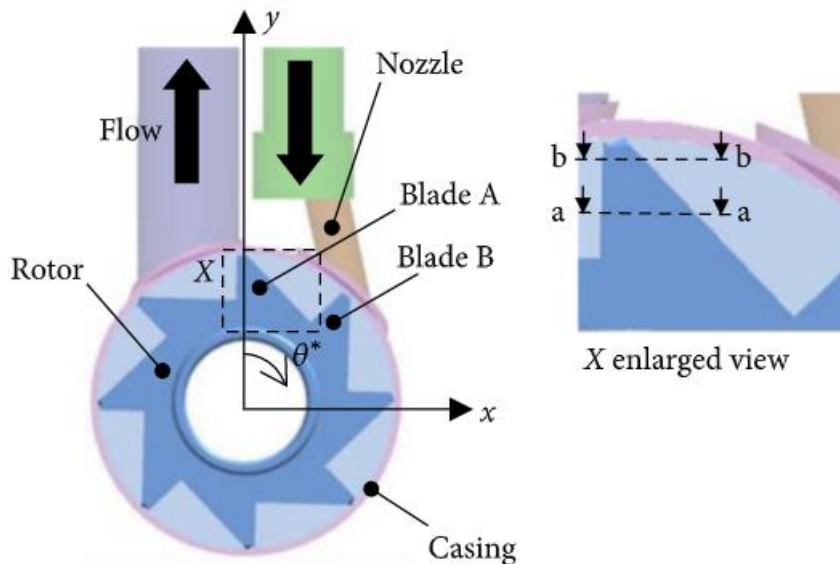


Figure 3.3.2 Turbine section region

For the coordinate system, we defined x and y -axes as in Figure 5. For the blade phase angle Θ^* , the position at which the front surface of the reference Blade A is parallel to the y -axis (the position in Figure 5) is set as $\Theta^* = 0^\circ$, and the counterclockwise direction was set as positive. Furthermore, we defined the blade ahead of Blade A as Blade B.

The cross-section at 3.7 mm from the rotation center in the perpendicular direction to the front surface of Blades A and B was defined as cross-section a–a, and the cross-section at 4.1 mm was defined as cross-section b–b.

3.4 Results and Discussion

3.4.1 Turbine Performance

Figure 3.4.1 shows the relation between T and rotational speed n , while Figure 3.4.2 shows the relation between turbine output L and rotational speed n . Figures 3.4.1 and 3.4.2 show that calculated T and L are slightly higher than the experimental values, but both T and L are consistent. Considering that calculated T and L ignore the loss of mechanical friction on the bearing, these analytical results are valid. Experimental and calculated T decreased with increase in rotational speed, and calculated L showed the maximum value at $n=250,000 \text{ min}^{-1}$ ($L = 12.1 \text{ W}$).

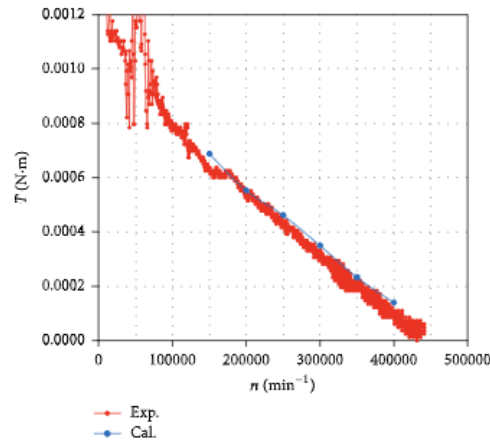


Figure 3.4.1 Correlation between rotational speed and torque

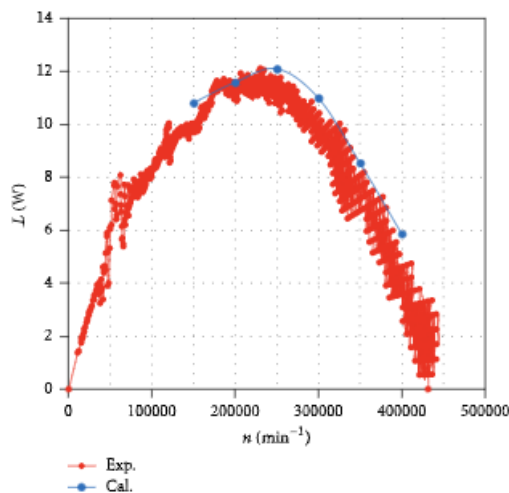


Figure 3.4.2 Correlation between rotational speed and turbine output

Next, to search the locations where loss inside of the hand-piece is high, we created a measurement surface in seven locations, as shown in Figure 11, and obtained entropy E for each measurement surface. Figure 12 shows the entropy E on each measurement surface for the hand-piece when the maximum output rotational speed was $n = 250,000 \text{ min}^{-1}$. Figure 12 shows that E demonstrated the highest increase in locations that are equivalent to the rotor region inside of F4-F5 casing among measurement surfaces. Therefore, in terms of the loss inside of the hand-piece, the loss in the rotor region is the most dominant. However, E increased slightly in F1–F4 inlet flow and F6–F7 outlet flow; thus, the loss in the inlet and outlet flows cannot be ignored.

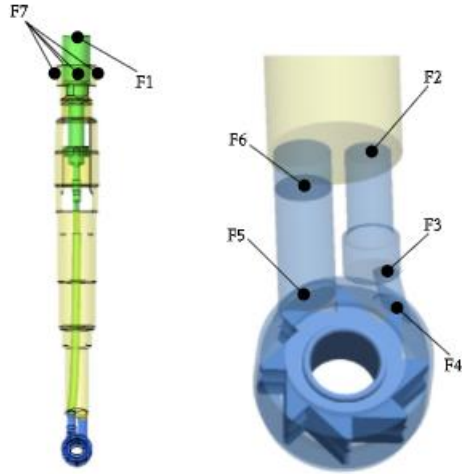


Figure 3.4.3 Measuring surface

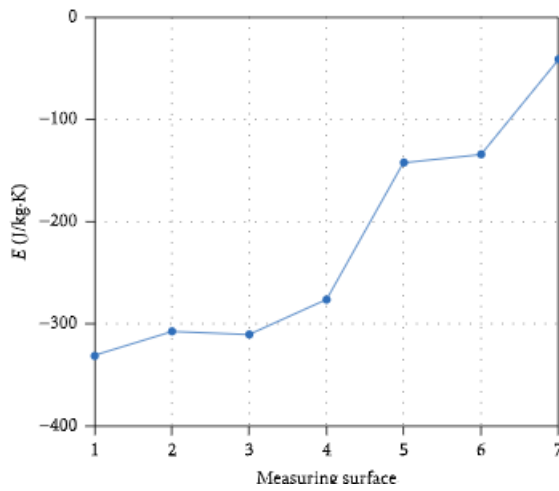


Figure 3.4.4 Entropy at each measuring surface.

We obtained adiabatic efficiency η_{ad} of the turbine section and the whole hand-piece with the following equations:

$$\eta_{ad} = \frac{L}{L_{ad}}, \quad (2)$$

$$L_{ad} = \dot{m} \left[\frac{\kappa}{\kappa - 1} \frac{P_1}{\rho_1} \left\{ 1 - \left(\frac{P_2}{P_1} \right)^{(\kappa-1)/\kappa} \right\} + \frac{1}{2} (v_1^2 - v_2^2) \right],$$

Where m is the mass flow rate and k is the specific heat ratio (1.4). ρ_1 , p_1 , and v_1 are fluid density, static pressure, and absolute velocity at the inlet, respectively. To calculate η_{ab} of the turbine section, we used the value from the turbine section inlet (F2); to calculate η_{ab} of the whole hand-piece, we used the value from the hand-piece inlet (F1). p_2 and v_2 are static pressure and absolute velocity at the outlet, respectively, and to calculate η_{ab} of the turbine section we used the value from the turbine section outlet (F6), and to calculate of the whole hand-piece, we used the value from the hand-piece outlet (F7).

Figure 13 shows the relation between adiabatic efficiency η_{ab} and rotational speed n of the turbine section and the whole hand-piece. Figure 13 shows that η_{ab} of the turbine section and the whole hand-piece present the maximum value when $n=250,000 \text{ min}^{-1}$, which is similar to the maximum value of the turbine output, and they show a major drop at high rotational speed. The maximum η_{ab} of the turbine section was quite low at 0.185. The maximum η_{ab} of the whole hand-piece was 0.086, and this is approximately 46% less than the maximum η_{ab} of the turbine section.

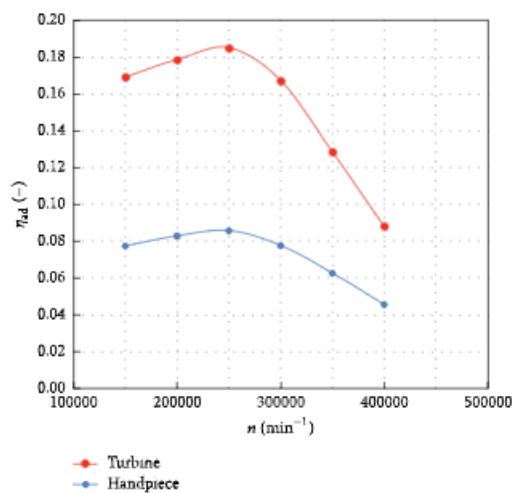


Figure 3.4.5 Correlation between speed and adiabatic efficiency

3.4.2 Torque Fluctuation and Internal Flow

Figure shows the torque fluctuation during one rotor rotation for each rotational speed n . Figure 14 shows that T fluctuates greatly during one rotor rotation, showing eight peaks in response to the number of blades. Fluctuation in T increases with increased rotational speed, and especially at $n=400,000 \text{ min}^{-1}$, the minimum value presents a negative value.

When torque fluctuation increases, it lowers workability of the air turbine headpiece and causes fluid noise; thus, suppression of these issues is a challenge. Since torque fluctuates, periodically to the number of blades, torque fluctuation for one blade pitch is shown in Figure 3.4.2[1]. Figure 3.4.2[1] shows that when the blade phase angle is around $\Theta^* = 36^\circ$, T becomes a minimum. Subsequently, T increases rapidly and after reaching the maximum around $\Theta^* = 48^\circ$, it gradually decreases.

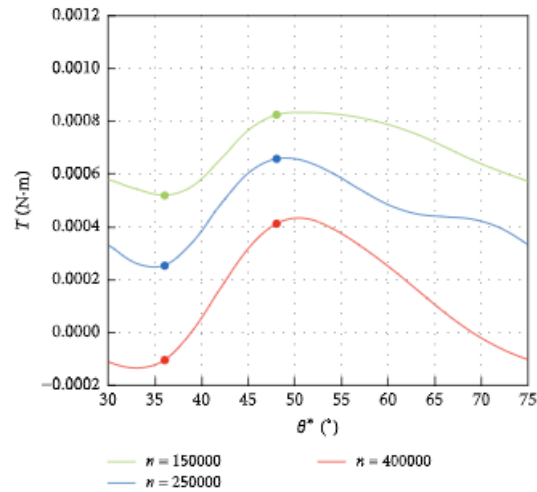


Figure 3.4.2[1] Torque Fluctuation for one blade pitch.

Therefore, we focused on these blade phase angles and examined the internal flow. First, we present static pressure distribution and absolute velocity vectors for the center cross-section of the blade width for each blade phase angle ($\Theta^* = 36^{\circ}, 48^{\circ}$) at the maximum output rotational speed of $n = 250,000 \text{ min}^{-1}$ (Figures 16(a) and 16(b)). Figures 17-24 show static pressure distribution and absolute velocity vectors of cross-sections a – a and b–b for each blade (Blade A and Blade B).

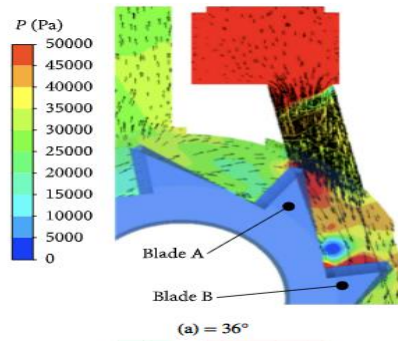


Figure. 3.4.2[2] Static pressure distribution and absolute velocity vectors ($n=250,000 \text{ min}^{-1}$)

CHAPTER 4:- RESULT AND ANALYSIS

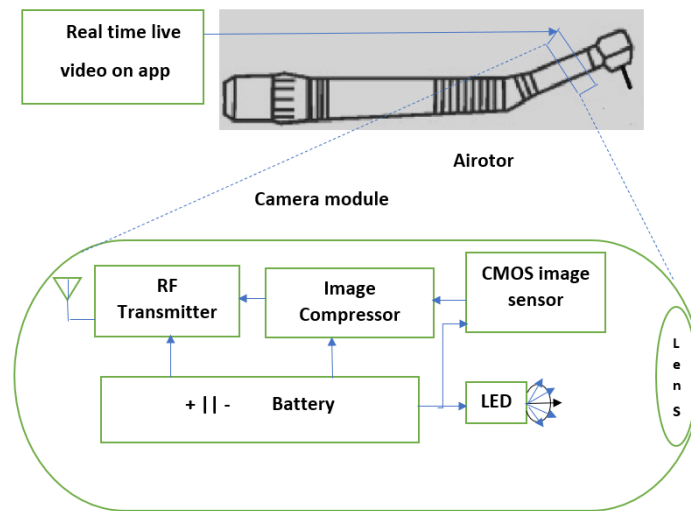


Figure 4.1 Block Diagram



Figure 4.2 Airotor



Figure 4.3 Endoscopic Camera



Figure 4.4 Simple Airotor



Figure 4.2 Airotor with Micro-camera attached

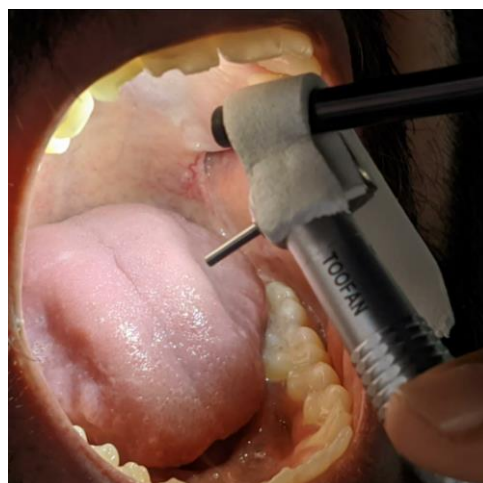


Figure 4.3 Airotor Demonstration

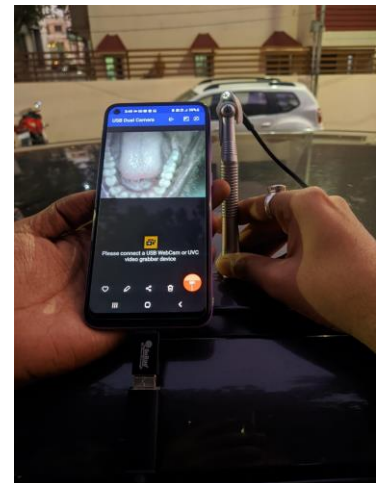


Figure 4.4 Final result with dual USB Application

CHAPTER 5:- CONCLUSION

1. From above process we have solved a problem which is widely encountered in dental Treatment.
2. From all of the above outcomes we have successfully managed to ensure the safety of The patient while performing the dental operations.
3. A treatment method by a hand-piece with a built-in camera, comprising:
Performing a treatment while viewing a clear color image of a treatment site, which is Difficult to be seen normally, such as a distal wall of a tooth, by using the hand-piece With a built-in camera.
4. Wireless communication means for allowing an image signal imaged by the camera unit To be wirelessly communicated with the outside.
- 5.

CHAPTER 6:- FUTURE SCOPE

1. With the help of micro-camera based airotor we will be able to fit camera in hand-piece....so it will enable The possibilities of using machine learning capabilities to detect infections in cavity.
2. An anti-fogging mechanism which blows air toward a front side of the light receiving port can Also be added to take the high resolution photos while performing operations itself while using Water gun of airotor.
3. In future there will be possibility of modular camera which will be attachable & detachable to Hand-piece / airotor

CHAPTER 7:- REFERENCES

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CHAPTER 8:- LIST OF PUBLICATION

Sr. No.	Authors	Title of Paper	Name of International Journals / International Conference	Place and date of Publication with Citation Index
1	Sushant Deo Bhushan Khandekar Himanshi Tekade Aishwarya Petkar Rutuja Jambhulkar	MICRO-CAMERA BASED AIROTOR	International Conference For Dental Reserarch Presentation 2021 (ICDRP 2021)	October/November 2021 Organized by VSPM's Dental College & Research Centre, Nashik.

Table 2. Publication

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